Static Voltage Stability Analysis and Control Strategy of Distribution Network with High Proportion Distributed PV Generation during Fault

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2025 International Conference on New Power System Technology (PowerCon) | 979-8-3315-6552-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/POWERCON66300.2025.11295112

***Abstract*—To address the issue of static voltage stability in distribution networks with high proportion distributed pv generation(DG) during fault, this paper establish a fault equivalent model which consider the characteristics of PV equipment during low-voltage ride-through(LVRT). The model derives the constraints on the PV current phase angle required to maintain static voltage stability. Subsequently, the impact of DG capacity at the point of common coupling and the electrical distance between the distribution network and the AC power source on the PV output current phase angle constraint is quantitatively analyzed. On this basis, a current control method is proposed: when there is a voltage stability risk in the system, the active and reactive current injection from the PV is adjusted, while under no voltage stability risk, the low-voltage ride-through control strategy remains unaffected. Finally, simulation examples based on the Matlab/Simulink testing platform verify the validity of the analysis and the proposed control strategy.**

Keywords—Distributed PV generation;Distribution networks; Static voltage stability

I. INTRODUCTION

The sustainable development of DG contributes to promoting energy conservation, emission reduction, achieving sustainable economic and social development [1]. However, the rapid growth of DG is accompanied by increased operational risks to distribution networks and voltage stability issues arise during low-voltage ride-through (LVRT) in certain areas. Researching the voltage stability mechanism and control strategies of DG during LVRT has a great significance for improving power quality and supply reliability.

Currently, the main methods which is short circuit ratio [2][3], eigenvalue analysis [4], and impedance margin [5] is used to evaluate the static voltage stability of power systems. Reference [6] considers the impact of different power growth modes on the static voltage stability of the system. This academic proposes the short circuit ratio index for new energy station. Reference [7] establishes a bi-level optimal power flow model to determine the static voltage stability margin interval, which considering the uncertainty of renewable energy stations. Reference [8] bases on the characteristics of radial distribution systems. Then the academic proposes a method a voltage stability analysis which is suitable for distribution networks. Reference [9] uses the singularity of the power flow Jacobian matrix at the static voltage stability critical point, and describes this characteristic by setting the determinant of the Jacobian to zero [10].

During LVRT, the output current of DG is "Both magnitude and phase are constrained." Once integrated into distribution networks, the traditional short circuit ratio and impedance margin methods cannot fully consider the characteristics of PV devices [12], which makes it difficult to define a clear boundary value for the corresponding results [13]. The eigenvalue evaluation method requires computing the eigenvalues of the Jacobian matrix, which entails substantial computational effort and low online calculation efficiency, making it unsuitable for online stability control applications [14]. Stability control based on constructing dynamic compensation control loops is more readily implemented which apply on a large number of DG in distribution networks [15]. Current research mainly focuses on transient stability of power angle and frequency, while research on this aspect is relatively limited [16].

This paper analyzes the static voltage stability mechanism of DG during LVRT, and derives the range of DG current output angles that ensure voltage stability in the distribution network. Based on this, the paper further examines the influence of DG's capacity and the electrical distance ,which is between distribution network and AC power sources , on the range of PV output current angles. Subsequently, the critical current value for voltage stability is derived, and a current critical value calculation module is designed. When the classic LVRT strategy of DG would otherwise lead to voltage instability, the current calculation module adjusts the DG output current dynamically. Finally, simulation cases built in MATLAB/Simulink are used to verify the effectiveness of the proposed strategy.

II. ANALYSIS OF STATIC VOLTAGE STABILITY OF DG DURING LOW-VOLTAGE RIDE-THROUGH

A. Mechanism Analysis of Static Voltage Stability of DG During LVRT

The fault scenario model of a distribution network with DG is shown in Fig.1. DG_1 and DG_2 connected at grid-connection points 1 and 2, respectively. Loads 1 and 2 are equivalent loads at the respective grid-connection points. Assuming that a symmetrical ground fault occurs on the 10 kV line between grid-connection points 1 and 2, with R_f

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representing the transition resistance. DG_1 connected to the main grid via transformer, which is defined as main-grid-side PV. DG_1 connected a 10 kV line with no synchronous power source support, which is defined as distribution-network-side. This paper analyzes the static voltage stability mechanism of main-grid-side PV.

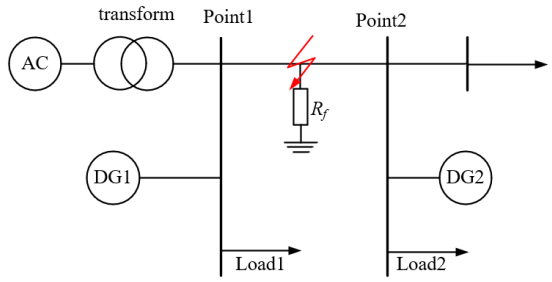


Fig.1 Distributed generator connected to the distribution network fault diagram

Considering the mechanism of static voltage stability during LVRT of DG_1. The equivalent fault model is shown in Fig.2. In the figure, E represents the equivalent synchronous power source, Z_s is the equivalent impedance between the synchronous power source and the grid-connection point of DG_1. $U_1\angle\Phi$ denotes the magnitude and phase angle of the point1, $I_{1max}\angle\Phi$-$\Delta\Phi$ represents the equivalent controlled current source of DG_1, where $\Delta\Phi$ is the phase angle which the PV current lags the voltage. $Z_{1l}\angle\Theta_l$ is the line impedance from the grid-connection point to the fault location, $U_f\angle\Theta_f$ represents the magnitude and phase angle of the feeder voltage at the fault point. $Z_{load}\angle\Theta$ is the equivalent load impedance.

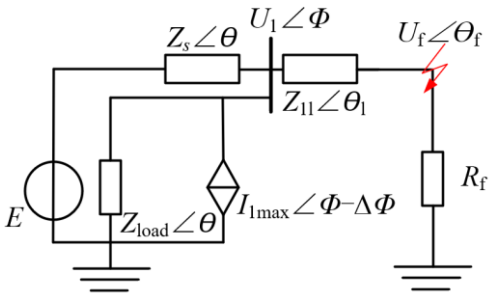


Fig.2 Equivalent Model During Fault on the DG1 Side

According to Fig.2, the electrical relationship between the voltage at grid-connection point 1 and the fault point can be expressed as follows:

$$U_1\angle\phi-\left(\frac{E}{Z_s\angle\theta}+I_{1max}\angle(\phi-\Delta\phi)-\frac{U_1\angle\phi}{(Z_s+Z_{load})\angle\theta}\right)Z_{1l}\angle\theta=U_f\angle\theta_f \quad (1)$$

By taking the modulus of equation (1) and solving, the expression for the fault point voltage can be obtained as follows:

$$\begin{cases} U_f=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a} \\ a=1 \\ b=-2\left[\left(1+\dfrac{Z_{1l}}{(Z_s+Z_{load})}\right)U_1\cos(\phi-\theta_f)+Z_{1l}I_{1max}\cos(\theta-\Delta\phi)\right] \\ c=-2\left(1+\dfrac{Z_{1l}}{(Z_s+Z_{load})}\right)Z_{1l}I_{1max}\cos(\theta-\Delta\phi)U_1+Z_{1l}^2I_{1max}^2+\left(1+\dfrac{Z_{1l}}{(Z_s+Z_{load})}U_1\right)^2-\left(\dfrac{Z_l}{Z_s}E\right)^2 \end{cases} \quad (2)$$

Equation (2) gives the expression for the voltage magnitude at the fault point. The condition for a feasible solution is b2-4ac>0. Furthermore, the voltage magnitude cannot be negative, so there must be at least one positive solution. According to the fault point voltage expression in equation(2), the non-square-root part of the solution, b/2a, is:

$$\left(1+\frac{Z_{1l}}{(Z_s+Z_{load})}\right)U_1\cos(\phi-\theta_f)+Z_{1l}I_{1max}\cos(\theta-\Delta\phi) \quad (3)$$

Θ represents the impedance angle of equivalent load of the transformer area, with its value in the range Θϵ[00,900]. Therefore, the difference between the PV current lag angle and the impedance angle, (ΔΦ-Θ) , lies within[-900,900], and thus cos(ΔΦ-Θ) is always greater than zero. (Φ-Θf) denotes the angle between the grid-connection point and the fault point. Since the transition resistance is generally purely resistive, the voltage phase at the fault point is the same as the phase of the short-circuit current, so (Φ-Θf) is equal to the phase angle difference between the voltage and current at the grid-connection point; cos(Φ-Θf) is always greater than zero. Therefore, equation(3) is always positive, indicating that there will always be at least one positive solution for the fault point voltage as long as a feasible solution exists.

By combining equation(2) to solve b2-4ac>0, the current phase angle of DG1 output current must satisfy the following condition for the fault point voltage to be solvable:

$$\left|\sin(-\theta-\phi+\Delta\phi)+\frac{\left(1+\dfrac{Z_{1l}}{(Z_s+Z_{load})}\right)U_1\sin(\phi-\theta_f)}{Z_{1l}I_{1max}}\right|<\left|\frac{E}{Z_sI_{1max}}\right| \quad (4)$$

B. Impact Analysis of LVRT on Static Voltage Stability of Distribution Networks

The proportion of active and reactive components in the PV output current can be represented by the ratio of i_d to i_q . The corresponding phase angle can be expressed as follows:

$$\Delta\phi=\arcsin\frac{i_q}{I_{max}} \quad (5)$$

Combining the LVRT control commands, the following can be obtained:

$$\Delta\phi=\arcsin\frac{KI_N(0.9-U)}{I_{max}} \quad (6)$$

Taking the stability criterion on the main grid side as an example, according to equation (4), it can be seen that the maximum DG integration capacity Imax and the distance parameter between the distribution network and the synchronous power source, Zs, will affect static voltage stability. By setting Zs=0.3p.u, the relationship between the actual current angle and the stable operating region of the distribution network under different PV capacities is illustrated, as shown in Fig.5. Similarly, by setting Is=0.8p.u, the relationship between the actual current angle and the stable operating region under different parameters of the distribution network and synchronous power source is shown in Fig. 5 and Fig. 6.

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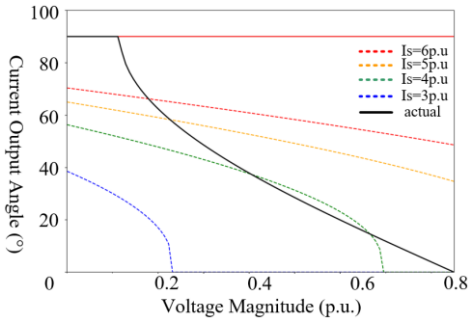


Fig.5 Operating stability region under different DG1 penetration levels

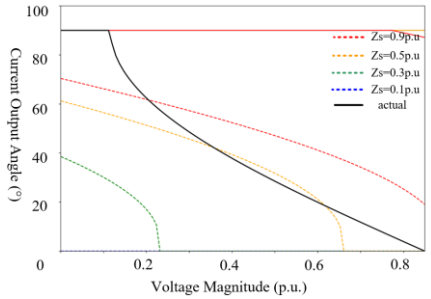


Fig.6 DG1 Operating stability region under different Zs

III. Active Voltage Control Strategy for DG during LVRT

The requirement for the existence of a feasible solution for the DG_1 is that the output current angle of the PV satisfies equation(4). By combining this with equation(5), the lower limits of the q-axis and d-axis currents can be obtained as follows:

$$\begin{cases} i_{\text{qmin}} = I_{\text{max}} \sin \Delta\phi_{\text{min}} \\ i_{\text{dmin}} = I_{\text{max}} \cos \Delta\phi_{\text{max}} \end{cases} \tag{7}$$

The upper limit of the d-axis current can be calculated using the lower limit of the q-axis current and the maximum output capacity of the PV system. The reference calculation formula can be expressed as follows:

$$\begin{cases} i_{\text{qmin}} = I_{\text{max}} \sin \Delta\phi_{\text{min}} \\ i_{\text{dmax}} = \sqrt{I_{\text{max}}^2 - i_{\text{qmin}}^2} \end{cases} \tag{8}$$

Based on equation(8), a calculation module for limiting the current loop reference value is constructed. The current control structure is designed as shown in the block diagram in Fig.7 and Fig.8.

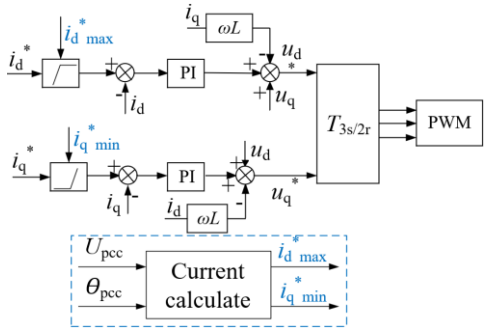


Fig.7 Current limit module connection position

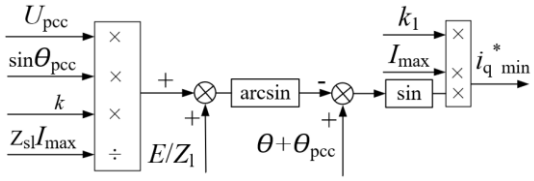


Fig.8 Diagram of current limit value calculation

IV. Simulation Verification

A distribution network model ,which structure refers to Fig.1. is established by MATLAB/Simulink. The parameters bases on an actual system in Jiangsu, China. To verify the effectiveness of the proposed strategy, we compare the voltage stability capability when the PV adopts the conventional LVRT and the control strategy proposed in this paper while short-circuit faults on the distribution lines under different scenarios.

A. Simulation under different transition resistance scenarios

The I_s is set to 3 p.u, and the Z_s is set to 0.9 p.u A symmetrical grounding fault with a transition resistance of 15 Ω is applied at the midpoint between point 1 and point 2. During the fault, the value of the transition resistance is gradually increased, and the variation of the PV's reactive power output on the main grid side is recorded. The main grid side reactive power output is shown in Fig. 9.

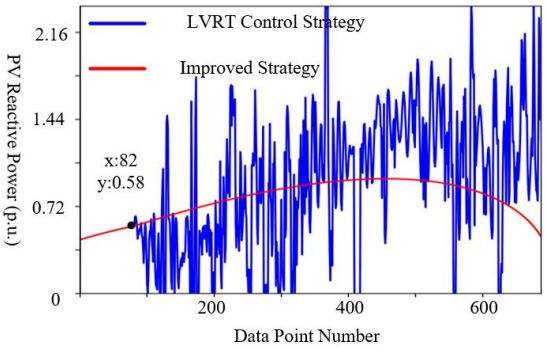


Fig.9 DG reactive power which apply LVRT and improved strategy under different Rf

When the transition resistance reaches a critical point, PV adopting the low-voltage ride-through (LVRT) control strategy loses static voltage stability. By contrast, the improved control strategy limits the current reference value, enabling the PV to increase reactive power output and reduce active power output, thereby maintaining stable system operation. By calculating the terminal voltage using equation (8), the corresponding voltage is 0.21p.u. The upper and lower bounds of the PV current constraint and actual current output angle under different voltage are determined, while I_s = 3p.u and Z_s = 0.9p.u, as shown in Fig.10. When the voltage sag is 0.232 p.u., the critical point of static voltage instability coincides with the intersection of the PV current output angle and the lower bound of the stability region. When the voltage continues increase, voltage stability issues arise. These results theoretically verify the reliability of the proposed strategy.

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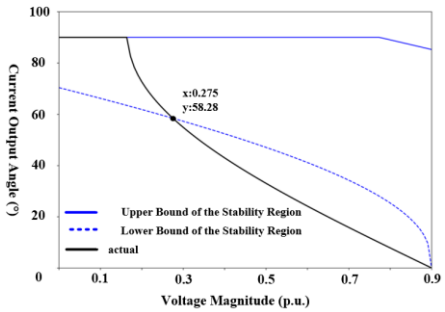
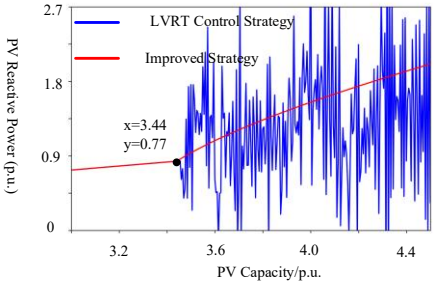


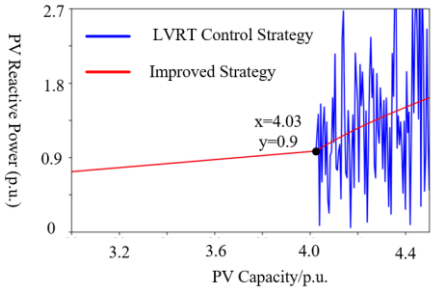
Fig.10 PV stable operating region(Is=0.8 p.u.,Zs=0.3 p.u.)

B. Simulation Results for Various Z_s

To verify the effectiveness of the improved strategy under different electrical distances, two scenarios with electrical distances Zs of 0.3 p.u and 0.5 p.u are set in this study. During the fault ride-through period, the PV output is gradually increased until voltage instability occurs, with the voltage set to 0.6 p.u. The simulation results are shown in Fig. 11.



(a) Zs=0.5 p.u.



(b) Zs=0.3 p.u.

Fig.11 DG reactive power which apply LVRT and improved strategy under different Zs

As shown in Fig.11(a), when the PV capacity is increased to 3.44p.u, the PV system under LVRT control loses static voltage stability. However, with the improved control strategy, the PV can maintain voltage stability. As shown in Fig. 11(b), when the electrical distance to the main grid is reduced to 0.3p.u, the critical point of voltage instability occurs when the PV output current rises to 4.03p.u. Compared to the scenario with Zs=0.5p.u., the distribution network can accommodate a higher PV output current, indicating that a shorter electrical distance increases the network's capacity for PV output.

At the critical point of voltage instability under the LVRT control, the output power is recorded, and the corresponding PV current output angle is calculated using equation(10), resulting in a value of 23.9°. Using the previously described method for analyzing the electrical distance, the lower boundary of the current output angle for stable operation and actual current output angle are determined, which is shown in Fig. 13. When Is=3.44p.u., the lower limit of the PV current output angle for stable voltage operation is less than 23.9° in the Zs=0.5 p.u. case.

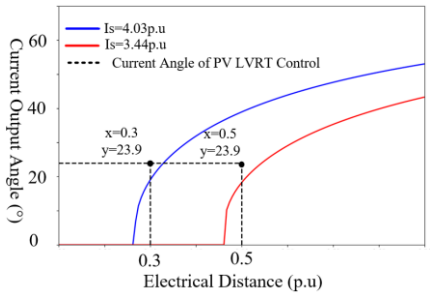


Figure 12. Lower Boundaries of Voltage Stability Region and Output Current Angles at Different Main–Distribution Electrical Distances

V. Conclusion

To address the problem of system voltage instability caused by photovoltaic current injection during faults, this paper proposes a static voltage stability analysis and control strategy for distribution networks during faults. The following conclusions are obtained through simulation verification:

(1) The large access capacity of distributed photovoltaic power and the long electrical distance between the distribution network and the AC power supply are the main factors leading to static voltage instability during faults.

(2) The strategy proposed in this paper calculates the output Angle range of the photovoltaic current under the condition of voltage stability during the fault period, and then corrects the current reference value of the low-penetration strategy, which can effectively improve the static voltage stability during the fault ride-through period.

Compared with the classic low-voltage ride-through control strategy, the strategy proposed in this paper has a smaller computational load and does not require significant modifications to the photovoltaic control structure, thus having stronger applicability.

Acknowledgment

This work was financially supported by County-level distributed power source regulation simulation and auxiliary decision-making system(B442NY240013).

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